

When Does Validated Memory Transfer in LLM Agents? Evidence from Embodied, SQL, and Science Domains

Anonymous ACL submission

Abstract

LLM-based agents are increasingly augmented with episodic memory, but stored experience only helps when past corrections transfer to future tasks. We study a validated-memory framework that extracts outcome-backed issue-learning pairs from Reflexion trajectories and retrieves them through proactive *Context Retrieval* (CR) and on-demand *Tool Retrieval* (TR). Across final audited runs in ALFWorld, SQL, and ScienceWorld, memory effects are sharply conditional. In ALFWorld, a transfer-friendly embodied text environment, CR+TR improves ReAct from 67.9% to 82.8% success (+14.9pp), while hard-negative retrieval falls to 55.2%, supporting retrieval relevance as causal rather than extra context alone. In SQL, CR+TR yields only a narrow gain over ReAct (92.0% vs. 91.5%), and CR alone plus hard-negative retrieval hurt. In ScienceWorld, ReAct remains best in the final audit (30.0%), and a later 3-variation/category density gate still fails to promote CR+TR. We argue that validated memory can improve transfer when reusable procedures and meaningful retrieval relevance align, but memory density alone is insufficient; transfer radius helps explain the positive, mixed, and negative cases.

1 Introduction

LLM-based agents are increasingly augmented with memory systems that store prior experience and retrieve it during future tasks. This design is appealing: if an agent has already corrected a failure once, the correction should be reusable later. Yet this premise hides a central question: *when does validated memory actually transfer?*

We study this question using a single memory framework across three final audited settings: ALFWorld (Shridhar et al., 2021), SQL query generation, and ScienceWorld (Wang et al., 2022). The framework extracts issue-learning pairs from successful Reflexion (Shinn et al., 2023) training trajectories, retains validation metadata, and injects

those memories at inference time through Context Retrieval (CR), Tool Retrieval (TR), or both. All final evaluations use the same base ReAct (Yao et al., 2023) agent, Gemini 2.5 Flash, train-only knowledge bases, and hard-negative retrieval controls.

The results are conditional rather than universal. ALFWorld is the clean positive case: CR+TR improves success from 67.9% to 82.8% on 134 held-out validation-unseen tasks, a +14.9pp gain. The hard-negative control falls to 55.2%, which supports the interpretation that relevant retrieved memory is doing useful work rather than merely adding more prompt text. SQL is a narrow-transfer diagnostic: CR+TR reaches 92.0% success versus 91.5% for ReAct, while CR alone and hard-negative retrieval both hurt. ScienceWorld is a diagnostic negative case: ReAct remains best in the final audit at 30.0% success, CR+TR averages 27.8%, and hard-negative CR+TR matches or exceeds CR+TR on the headline success metric.

These outcomes motivate a transfer-radius view of memory-augmented agents. Validated memories help when environments contain reusable procedures and when retrieval relevance aligns with those reusable structures. They are weak or harmful when retrieved memories are too narrow, when task structure changes faster than the memory can generalize, or when increased memory density does not improve retrieval alignment. A post-hoc but pre-registered ScienceWorld density gate strengthens this interpretation: expanding the ScienceWorld training knowledge base to three variations per category improved some raw numbers, but still did not promote ScienceWorld because CR+TR underperformed same-run ReAct on mean success and reward, while hard-negative retrieval matched or beat CR+TR on success.

In summary, our contributions are:

1. A **final audited evaluation** of validated episodic memory across ALFWorld, SQL, and

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ScienceWorld using train-only trials, knowledge bases, held-out evaluation, and hard-negative controls.

2. **Evidence for conditional transfer:** a strong ALFWorld gain (+14.9pp), a narrow SQL gain (+0.5pp), and a ScienceWorld non-promotion where memory density alone is insufficient.
3. A **diagnostic framing** based on task compositionality, action-space regularity, and learning transfer radius, explaining when CR and TR should be expected to help, fail to add, or mislead.

2 Related Work

Tool-augmented and memory-augmented agents. ReAct (Yao et al., 2023) interleaves reasoning with environment actions, and Reflexion (Shinn et al., 2023) extends this pattern with iterative self-correction. Several systems augment LLM agents with persistent memory or reusable skills: Voyager (Wang et al., 2023) builds a verified skill library, GITM (Zhu et al., 2023) stores goal decompositions, Generative Agents (Park et al., 2023) use a memory stream with reflection, MemoryBank (Zhong et al., 2024) incorporates forgetting curves, A-Mem (Xu et al., 2025) treats memory as an agentic module, and EvoSkill (Alzubi et al., 2026) evolves reusable skills from failures. Our work studies a complementary question: not whether memory can help in favorable domains, but which final audited settings show transferable benefit under matched retrieval and hard-negative controls.

Retrieval-augmented generation and agent retrieval. Retrieval-augmented generation conditions model outputs on external documents or trajectories (Lewis et al., 2020). In agent settings, trajectory retrieval can provide demonstrations, warnings, or issue-specific guidance. However, high semantic similarity is not identical to transfer utility. We therefore include hard-negative controls that preserve the presence of retrieved content while inverting relevance, allowing us to distinguish useful memory from generic prompt expansion.

Interactive evaluation domains. ALFWorld (Shridhar et al., 2021) exposes structured household tasks with repeated subgoals and

a constrained text action grammar. ScienceWorld (Wang et al., 2022) exposes scientific procedures across many task categories, with richer dynamics and more varied validation signals. SQL query tasks provide a high-accuracy but narrow-transfer setting: many examples share syntax and schema-manipulation patterns, yet individual memories can be brittle when they encode local query choices. This mix lets us test memory transfer across a positive case, a narrow diagnostic case, and a diagnostic negative case without relying on exploratory or superseded runs as final evidence.

3 Methodology

We evaluate a unified validated-memory framework across three final audited settings with different transfer structure.

3.1 Environments

ALFWorld. ALFWorld (Shridhar et al., 2021) provides a text-based interface to the ALFRED embodied household benchmark. Agents issue actions from a fixed grammar of ~ 15 verb templates (e.g., go to, take, open, heat). Six task types share a compositional goal structure: SEARCH $\rightarrow$ ACQUIRE $\rightarrow$ TRANSFORM $\rightarrow$ PLACE. The small object vocabulary (~ 50 types) and shared sub-goal phases yield high structural overlap between tasks.

SQL. The SQL setting asks agents to produce executable queries for held-out database tasks. It has a regular symbolic action language and many reusable syntactic patterns, but individual memories often have narrow transfer radius: a useful query correction can depend on local schema names, joins, or predicate choices that do not recur broadly.

ScienceWorld. ScienceWorld (Wang et al., 2022) contains science-themed interactive tasks spanning 30 categories. It exposes reusable procedural motifs, but the final KB contains only one train variation per category and the retrieval/validation signal is noisy relative to the diversity of held-out tasks. We therefore treat it as a diagnostic boundary case rather than as a positive transfer result.

3.2 Base Agent: ReAct

All agents build on ReAct (Yao et al., 2023), interleaving reasoning traces with actions. We use Gemini 2.5 Flash as the base LLM for all experiments.

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3.3 Memory Systems

Our framework augments ReAct with a persistent knowledge base and two retrieval mechanisms: proactive *Context Retrieval* (CR) at task start and on-demand *Tool Retrieval* (TR) during execution. We emphasize that Context Retrieval (CR) and Tool Retrieval (TR) address complementary aspects of transfer: CR supplies task-level guidance before acting, while TR enables action- or issue-level reuse when the agent asks for help. Their combination is designed to expose whether validated memories are useful before and during execution.

3.3.1 Knowledge Base Construction

The knowledge base is constructed offline from training trajectories using LLM-based extraction. For each task, we extract *issue-learning pairs*: structured records associating a mistake with its successful correction. Each entry includes task metadata, the issue-learning texts, a validation level reflecting downstream outcome (VALID_NEXT_TRIAL, VALID_SAME_TRIAL, or CANDIDATE), and provenance information. Final retrieval uses VALID_NEXT_TRIAL entries. We assign scalar validation weights ω to each level: $\omega_{\text{next}} = 1.0$, $\omega_{\text{same}} = 0.9$, $\omega_{\text{cand}} = 0.6$, used by both retrieval mechanisms to prioritize high-confidence learnings in the general algorithm. Full schema details are in Appendix B.

3.3.2 Context Retrieval (CR)

CR operates proactively: before the agent acts, it retrieves relevant past learnings based on embedding similarity (using `gemini-embedding-001`) between the new task description and historical tasks. The top- k ($k=5$) most similar tasks are identified, their associated learnings are pooled and scored by:

$$\text{Score}_{\text{CR}}(\ell) = \text{sim}(\mathbf{v}_q, \mathbf{v}_{t(\ell)}) \times \omega_\ell \quad (1)$$

where sim is cosine similarity and ω_ℓ is the validation weight.

Dynamic context sizing. Rather than retrieving a fixed number of learnings, CR adapts the context window to neighborhood complexity. Let L_{max} denote the maximum learning count among the top- k tasks. The retrieval quota is:

$$N_{\text{pick}} = \min(\lceil \alpha \cdot L_{\text{max}} \rceil, p) \quad (2)$$

where $\alpha = 1.5$ is a scaling factor and $p \in [5, 25]$ is a tunable cap. The top-scored learnings are formatted as issue-learning pairs and prepended to

the agent’s prompt. The full procedure is given in Algorithm 1 (Appendix A).

3.3.3 Tool Retrieval (TR)

TR provides on-demand retrieval exposed as an action `help["issue"]`. It uses hybrid dense-sparse retrieval: cosine similarity over issue embeddings captures semantic matches, while BM25 (Robertson and Zaragoza, 2009) over concatenated fields captures lexical matches. The candidate pool is the union of the top-100 results from each strategy. Scores are normalized to $[0, 1]$ and fused:

$$\text{Score}_{\text{TR}}(e) = (0.7 \cdot \cos_{\text{norm}} + 0.3 \cdot \text{BM25}_{\text{norm}}) \times \omega_e \quad (3)$$

The top- k ($k=3$) results are returned as issue-solution pairs. The full procedure is given in Algorithm 2 (Appendix A).

3.3.4 Hard Negative Control

To verify that gains stem from retrieval *relevance* rather than additional prompt context, we introduce a hard negative control that inverts the selection criterion: CR retrieves the *least* similar tasks and TR selects from the bottom of each retrieval strategy, producing prompts with the same number of learnings but from maximally irrelevant tasks.

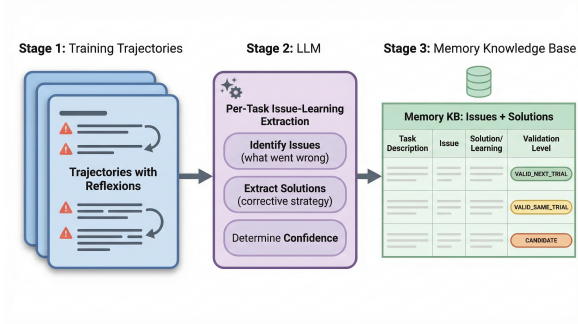
3.4 Reflexion for KB Construction

We use Reflexion (Shinn et al., 2023), an iterative self-correction framework, to generate training trajectories for the trials7 knowledge bases. Unlike inference-time CR and TR, Reflexion is intra-task: it retries the same task with accumulated self-critiques. Our final held-out comparisons evaluate single-trial ReAct variants with and without retrieved cross-task memory.

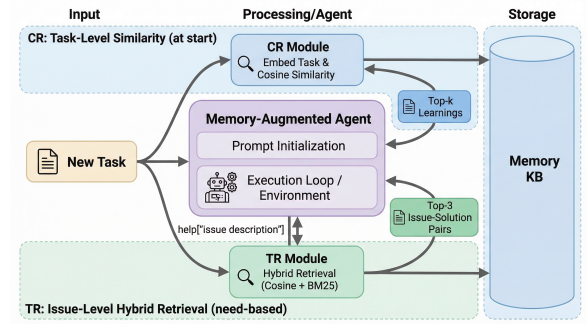
4 Experimental Setup

All agent reasoning uses Gemini 2.5 Flash, and all retrieval embeddings use Google’s `gemini-embedding-001`, held constant across experiments. The final Stage C audit passed with 0 errors and 0 warnings.

Tasks and splits. ALFWorld evaluation uses the standard validation-unseen split with 134 held-out tasks. SQL evaluation uses 200 held-out test tasks. ScienceWorld evaluation uses 30 held-out test categories. All final runs use train-only knowledge bases built from Reflexion trajectories with a maximum of seven trials. The final KB inventory is



(a) Offline KB construction from training trajectories.



(b) Inference-time retrieval via CR and TR.

Figure 1: Overview of the episodic memory framework. (a) Training trajectories with reflexions are processed by an LLM to extract per-task issue–learning pairs with validation levels. (b) Context Retrieval (CR) provides task-level learnings at the start; Tool Retrieval (TR) provides issue-level solutions on demand during execution.

shown in Table 2: ALFWorld has 373 memories, SQL has 415 sanitized memories, and ScienceWorld has 129 memories covering all 30 categories with one train variation per category.

Framework variants. We evaluate five inference-time variants per environment: ReAct, ReAct + CR, ReAct + TR, ReAct + CR + TR, and hard-negative CR + TR. All variants execute a single held-out trial per task. The hard-negative variant preserves memory injection while selecting low-relevance memories, making it a control for prompt length and generic extra context.

Metrics. We report success rate, score, gain from ReAct in percentage points, paired task-level deltas, retrieval counts, help-call counts, and average steps per task. For ALFWorld, the final ReAct rows did not emit reward, so success rate is used as the score in the aggregate table. For SQL and ScienceWorld, score is average reward. Provider dollar cost is not available in the final raw logs; the reward-vs-cost figure therefore uses average steps per task as a trajectory-cost proxy.

Audited artifacts. The final quantitative claims are anchored to the audited `final_runs/eval/` roots for ALFWorld, SQL, and ScienceWorld, plus their corresponding train-only KB roots under `final_runs/kb/`. The generated final tables and figures are included directly from the `paper/tables/final_*` and `paper/figures/final_*` artifacts.

5 Results

Table 1 reports the final held-out Stage C results. The headline is conditional: validated memory

helps strongly in ALFWorld, weakly in SQL, and fails to promote ScienceWorld.

5.1 ALFWorld: Positive Transfer

ALFWorld is the clean positive case. ReAct solves 67.9% of the 134 validation-unseen tasks, while CR+TR solves 82.8%, a +14.9pp improvement. TR alone is also positive (75.4%, +7.5pp), while CR alone is negative (63.9%, −4.0pp), showing that the two channels are useful but not automatically additive in isolation. The combined channel is strongest, suggesting that proactive task-level context and on-demand issue-level retrieval can complement each other when the environment has reusable procedures.

The hard-negative control is decisive in ALFWorld. Hard-negative CR+TR reaches only 55.2%, which is 27.6pp below proper CR+TR and 12.7pp below ReAct. This rules against a simple “extra context helps” explanation: retrieved memory helps only when it is relevant to the task structure.

5.2 SQL: Narrow Transfer

SQL is a narrow-transfer diagnostic rather than a large positive result. ReAct is already strong at 91.5% success, leaving little headroom. CR+TR improves to 92.0%, a small +0.5pp gain, but CR alone falls to 88.5% and TR alone to 90.8%. Hard-negative CR+TR reaches 89.5%, also below ReAct.

The SQL pattern indicates that some validated memories transfer, but the transfer radius is small. Useful query repairs can be highly local to a schema, predicate, or join pattern. This makes SQL evidence for conditional memory benefit: the best combined retrieval channel can help slightly, but individual channels and irrelevant retrievals can

Table 1: Final held-out Stage C results using train-only trials7 KBs, Gemini Flash, Gemini embeddings, VALID_NEXT_TRIAL, and max_learnings=5. Values are mean $\pm$ std over three seeds.

Environment	Framework	Tasks	Success (%)	Score	Δ success (pp)	
ALFWorld	ReAct	134	67.9 ± 0.0	0.679 ± 0.000	—	
ALFWorld	ReAct + CR	134	63.9 ± 1.6	0.639 ± 0.016	-4.0 ± 1.6	
ALFWorld	ReAct + TR	134	75.4 ± 0.0	0.754 ± 0.000	$+7.5 \pm 0.0$	
ALFWorld	ReAct + CR + TR	134	82.8 ± 0.0	0.828 ± 0.000	$+14.9 \pm 0.0$	
ALFWorld	Hard-neg CR + TR	134	55.2 ± 0.7	0.552 ± 0.007	-12.7 ± 0.7	
SQL	ReAct	200	91.5 ± 0.0	0.920 ± 0.000	—	
SQL	ReAct + CR	200	88.5 ± 0.0	0.889 ± 0.001	-3.0 ± 0.0	Score
SQL	ReAct + TR	200	90.8 ± 0.3	0.917 ± 0.003	-0.7 ± 0.3	
SQL	ReAct + CR + TR	200	92.0 ± 0.0	0.922 ± 0.000	$+0.5 \pm 0.0$	
SQL	Hard-neg CR + TR	200	89.5 ± 0.0	0.906 ± 0.000	-2.0 ± 0.0	
ScienceWorld	ReAct	30	30.0 ± 8.8	0.214 ± 0.241	—	
ScienceWorld	ReAct + CR	30	26.7 ± 3.3	0.171 ± 0.058	-3.3 ± 12.0	
ScienceWorld	ReAct + TR	30	26.7 ± 3.3	0.256 ± 0.086	-3.3 ± 6.7	
ScienceWorld	ReAct + CR + TR	30	27.8 ± 7.7	0.212 ± 0.134	-2.2 ± 6.9	
ScienceWorld	Hard-neg CR + TR	30	28.9 ± 6.9	0.260 ± 0.149	-1.1 ± 15.4	

is average reward when logged; for ALFWorld ReAct, success rate is used because reward was not emitted in the final record.

Table 2: Final train-only KB inventory for the trials7 protocol. VNT = VALID_NEXT_TRIAL.

Environment	KB artifact	Coverage	Max trial	Memories	VNT	Audit note
ALFWorld	alfworld_train reflexion_trials7	115/120 observed train games	7	373	101	Train-only refs; no eval overlap; original 120-game manifest absent.
SQL	sql_train_reflexion trials7_sanitized	516/516 train tasks	7	415	92	Sanitized; no dev/test overlap; protocol memories quarantined.
ScienceWorld	scienceworld_train reflexion_trials7 va80_30cat	30/30 categories, 30 variations	7	129	34	Train-only; 1 variation/category; failed extractions repaired.

easily perturb a high-performing baseline.

5.3 ScienceWorld: Diagnostic Negative Evidence

ScienceWorld should not be read as a positive transfer result. In the final Stage C audit, ReAct is the best framework at 30.0% success. CR, TR, and CR+TR all fall below ReAct on mean success, with CR+TR at 27.8%. Hard-negative CR+TR reaches 28.9%, matching or exceeding proper CR+TR on the headline metric.

This matters because the ScienceWorld final KB is clean but thin: it covers all 30 categories, yet only one train variation per category. The later Phase 4.1 density gate tested whether increasing to three train variations per category would rescue

the result. It did not: same-run CR+TR underperformed ReAct on mean success and mean reward, and hard-negative retrieval matched or beat CR+TR on success in every seed. We therefore use ScienceWorld as diagnostic evidence that memory density alone is insufficient when retrieval and validation signals do not align with reusable task structure.

5.4 Usage, Cost, and Paired Deltas

Table 3 shows that retrieval was active in the memory variants and that trajectory cost changes differ by environment. ALFWorld CR+TR uses both context memories and help calls while remaining far below the hard-negative variant in steps per task. SQL retrieval is sparse, especially for TR, consistent with its narrow-transfer character. Science-

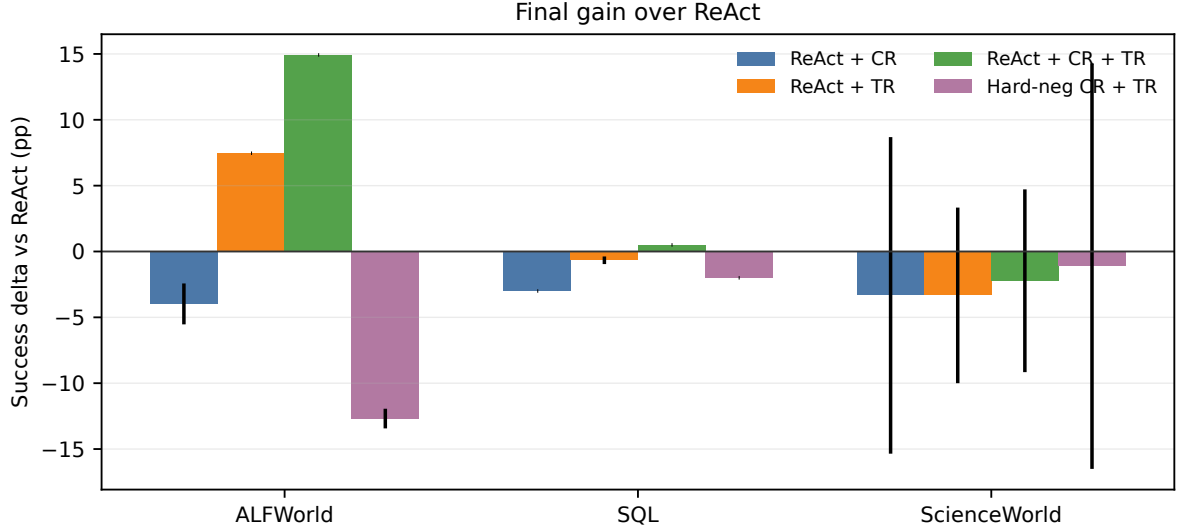


Figure 2: Final success gain over ReAct for each memory variant. ALFWorld shows a clear positive CR+TR effect, SQL shows a narrow CR+TR effect, and ScienceWorld does not show a promoted positive memory result.

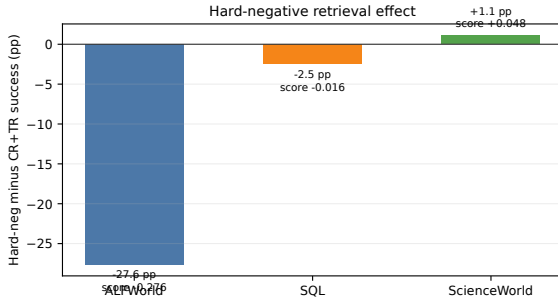


Figure 3: Hard-negative effect in the final audits. The ALFWorld gap between CR+TR and hard-negative retrieval supports relevance as causal; the SQL and ScienceWorld gaps are smaller or unfavorable.

World retrieval occurs reliably, but usage does not translate into better success.

Table 4 and Figure 4 sharpen the same picture at the task level. ALFWorld CR+TR improves many more tasks than it regresses. SQL CR+TR has only a small favorable imbalance. ScienceWorld has unstable paired deltas with no promoted CR+TR advantage.

6 Analysis and Discussion

The final results are best explained by a conditional transfer account. Validated memory is useful when stored learnings describe procedures that recur in held-out tasks and when retrieval relevance tracks that reuse. It is weak or harmful when memories are too local, when the action context changes, or when denser memory does not improve alignment.

6.1 Task Compositionality

ALFWorld tasks decompose into shared sub-goal phases such as searching, acquiring, transforming, and placing objects. A learning about opening a container, heating an object, or checking receptacles can apply across many held-out tasks. This compositionality makes cross-task memory useful.

SQL has compositional syntax, but many successful repairs are tied to local schema and predicate choices. ScienceWorld has procedural motifs, but its category diversity and sparse final KB make the mapping from retrieved learning to held-out task less reliable. The result is a gradient: strong ALFWorld transfer, narrow SQL transfer, and no promoted ScienceWorld transfer.

Principle. Episodic memory is most effective when tasks share reusable sub-procedures rather than merely superficial semantic similarity.

6.2 Action Space Regularity

ALFWorld’s fixed grammar makes action-level reuse tractable. TR is positive there, and CR+TR is strongest. SQL also has a regular symbolic language, but correctness depends on exact schema grounding; a memory can suggest the right kind of query while still pushing the agent toward the wrong table, join, or predicate. ScienceWorld actions and state transitions are more varied, and the final results show that successful retrieval volume is not enough to overcome this mismatch.

Table 3: Final memory usage and trajectory cost summaries from raw trajectories. Values are mean $\pm$ std over seeds.

Environment	Framework	Context tasks	Retrieved/task	Help calls	Help/task	Steps/task
ALFWorld	ReAct	0.0 $\pm$ 0.0	0.00 $\pm$ 0.00	0.0 $\pm$ 0.0	0.00 $\pm$ 0.00	21.00 $\pm$ 0.11
ALFWorld	ReAct + CR	132.0 $\pm$ 0.0	4.05 $\pm$ 0.00	0.0 $\pm$ 0.0	0.00 $\pm$ 0.00	28.56 $\pm$ 0.26
ALFWorld	ReAct + TR	0.0 $\pm$ 0.0	0.00 $\pm$ 0.00	302.3 $\pm$ 0.6	2.26 $\pm$ 0.00	27.69 $\pm$ 0.01
ALFWorld	ReAct + CR + TR	132.0 $\pm$ 0.0	4.05 $\pm$ 0.00	206.0 $\pm$ 0.0	1.54 $\pm$ 0.00	26.38 $\pm$ 0.01
ALFWorld	Hard-neg CR + TR	134.0 $\pm$ 0.0	4.67 $\pm$ 0.00	360.3 $\pm$ 6.4	2.69 $\pm$ 0.05	34.72 $\pm$ 0.08
SQL	ReAct	0.0 $\pm$ 0.0	0.00 $\pm$ 0.00	0.0 $\pm$ 0.0	0.00 $\pm$ 0.00	8.67 $\pm$ 0.00
SQL	ReAct + CR	178.0 $\pm$ 0.0	2.05 $\pm$ 0.00	0.0 $\pm$ 0.0	0.00 $\pm$ 0.00	8.94 $\pm$ 0.00
SQL	ReAct + TR	0.0 $\pm$ 0.0	0.00 $\pm$ 0.00	23.0 $\pm$ 0.0	0.12 $\pm$ 0.00	9.23 $\pm$ 0.00
SQL	ReAct + CR + TR	178.0 $\pm$ 0.0	2.05 $\pm$ 0.00	20.0 $\pm$ 0.0	0.10 $\pm$ 0.00	9.11 $\pm$ 0.01
SQL	Hard-neg CR + TR	171.7 $\pm$ 0.6	1.68 $\pm$ 0.01	24.0 $\pm$ 0.0	0.12 $\pm$ 0.00	9.03 $\pm$ 0.00
ScienceWorld	ReAct	0.0 $\pm$ 0.0	0.00 $\pm$ 0.00	0.0 $\pm$ 0.0	0.00 $\pm$ 0.00	30.96 $\pm$ 1.67
ScienceWorld	ReAct + CR	30.0 $\pm$ 0.0	4.40 $\pm$ 0.00	0.0 $\pm$ 0.0	0.00 $\pm$ 0.00	30.48 $\pm$ 1.24
ScienceWorld	ReAct + TR	0.0 $\pm$ 0.0	0.00 $\pm$ 0.00	37.3 $\pm$ 6.4	1.24 $\pm$ 0.21	34.19 $\pm$ 2.13
ScienceWorld	ReAct + CR + TR	30.0 $\pm$ 0.0	4.40 $\pm$ 0.00	20.7 $\pm$ 3.8	0.69 $\pm$ 0.13	31.23 $\pm$ 1.99
ScienceWorld	Hard-neg CR + TR	30.0 $\pm$ 0.0	4.17 $\pm$ 0.00	31.3 $\pm$ 2.1	1.04 $\pm$ 0.07	33.08 $\pm$ 0.37

Table 4: Paired task-level deltas against ReAct on the final held-out Stage C runs. Values are mean $\pm$ std over seeds.

Environment	Framework	Common tasks	Δ success (pp)	Δ score	Improved	Regressed
ALFWorld	ReAct + CR	134 $\pm$ 0	-4.0 $\pm$ 1.6	—	20.3 $\pm$ 1.5	25.7 $\pm$ 0.6
ALFWorld	ReAct + TR	134 $\pm$ 0	+7.5 $\pm$ 0.0	—	27.0 $\pm$ 0.0	17.0 $\pm$ 0.0
ALFWorld	ReAct + CR + TR	134 $\pm$ 0	+14.9 $\pm$ 0.0	—	33.0 $\pm$ 0.0	13.0 $\pm$ 0.0
ALFWorld	Hard-neg CR + TR	134 $\pm$ 0	-12.7 $\pm$ 0.7	—	18.7 $\pm$ 0.6	35.7 $\pm$ 0.6
SQL	ReAct + CR	200 $\pm$ 0	-3.0 $\pm$ 0.0	-0.031 $\pm$ 0.001	2.0 $\pm$ 0.0	8.0 $\pm$ 0.0
SQL	ReAct + TR	200 $\pm$ 0	-0.7 $\pm$ 0.3	-0.003 $\pm$ 0.003	2.0 $\pm$ 0.0	3.3 $\pm$ 0.6
SQL	ReAct + CR + TR	200 $\pm$ 0	+0.5 $\pm$ 0.0	0.002 $\pm$ 0.000	3.0 $\pm$ 0.0	2.0 $\pm$ 0.0
SQL	Hard-neg CR + TR	200 $\pm$ 0	-2.0 $\pm$ 0.0	-0.014 $\pm$ 0.000	2.0 $\pm$ 0.0	6.0 $\pm$ 0.0
ScienceWorld	ReAct + CR	30 $\pm$ 0	-3.3 $\pm$ 12.0	-0.043 $\pm$ 0.284	2.7 $\pm$ 1.2	3.7 $\pm$ 2.5
ScienceWorld	ReAct + TR	30 $\pm$ 0	-3.3 $\pm$ 6.7	0.042 $\pm$ 0.240	3.0 $\pm$ 1.0	4.0 $\pm$ 1.0
ScienceWorld	ReAct + CR + TR	30 $\pm$ 0	-2.2 $\pm$ 6.9	-0.002 $\pm$ 0.148	2.0 $\pm$ 1.0	2.7 $\pm$ 1.2
ScienceWorld	Hard-neg CR + TR	30 $\pm$ 0	-1.1 $\pm$ 15.4	0.046 $\pm$ 0.354	4.3 $\pm$ 1.5	4.7 $\pm$ 3.2

Principle. Regular action spaces help memory transfer only when the retrieved correction remains valid under the current task’s grounding constraints.

6.3 Learning Transfer Radius

We use *learning transfer radius* to describe how many future tasks can benefit from one stored learning. ALFWorld has high radius: one correction can apply across many object-location or transformation tasks. SQL has narrow radius: reusable syntax exists, but individual memories often depend on a particular schema or query intent. ScienceWorld has uncertain radius under the current retrieval setup: the 1-var final KB and 3-var density gate both fail to produce a reliable CR+TR advantage over ReAct or hard-negative retrieval.

Principle. Validated memory should be expected to help most when each stored learning has a broad transfer radius and retrieval can identify that radius before injecting the memory.

6.4 Why Hard-Negative Controls Matter

Hard-negative retrieval is essential because memory systems can appear to help for reasons unrelated to relevance: longer prompts, extra deliberation cues, or generic reminders. ALFWorld rejects that alternative explanation: proper CR+TR reaches 82.8%, while hard-negative CR+TR drops to 55.2%. SQL and ScienceWorld show the opposite cautionary value. When hard-negative retrieval is close to or better than proper retrieval, the memory channel has not demonstrated reliable relevance. This is why ScienceWorld remains diag-

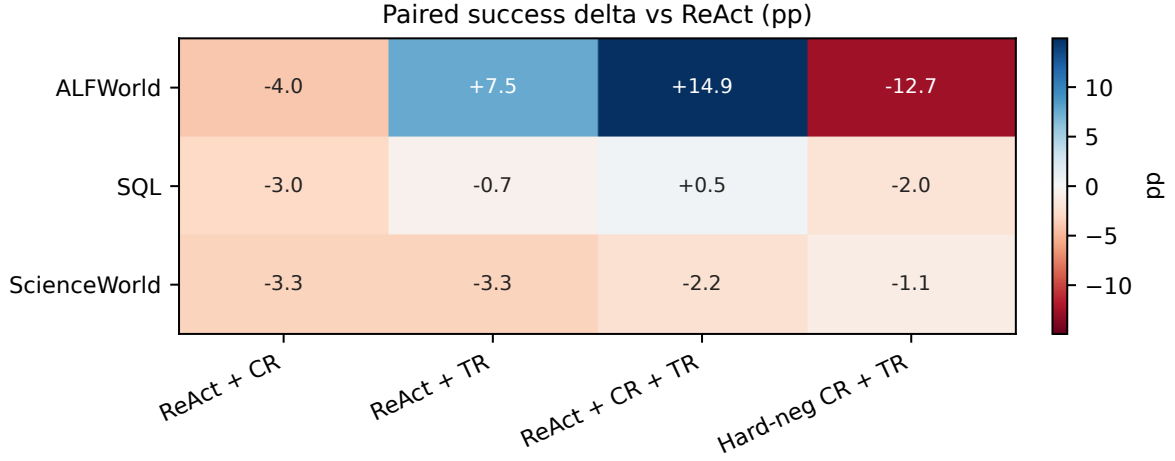


Figure 4: Paired task-level deltas against ReAct. Positive and negative cells expose where memory helps, regresses, or remains neutral under matched task comparisons.

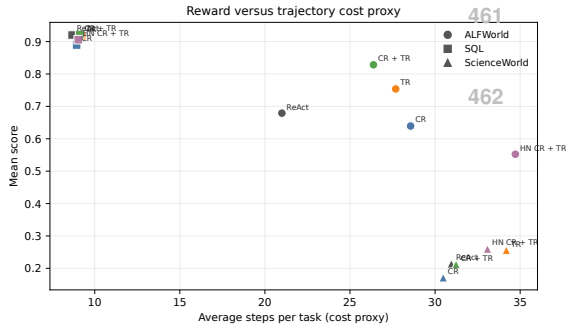


Figure 5: Final score versus trajectory cost, using average steps per task as a cost proxy because provider dollar cost is unavailable in the final raw logs.

7 Limitations

All final evaluations use one base LLM, Gemini 2.5 Flash. Different models may be more or less sensitive to retrieved memories, especially in narrow-transfer settings such as SQL or in procedural domains such as ScienceWorld.

The final evidence is intentionally restricted to audited runs. ALFWorld, SQL, and ScienceWorld are the only headline environments in the final tables. Older exploratory runs are preserved for provenance but are not used as final evidence here. This makes the claims cleaner, but it also means the transfer-radius framework should be treated as a diagnostic hypothesis requiring broader validation.

The knowledge bases are train-only, but they differ in density and provenance. The ALFWorld trials7 KB has no evaluation overlap after quarantine, but the exact legacy 120-game train manifest is absent and the audit observes 115 unique train trajectory task IDs. The ScienceWorld final KB covers all 30 categories with only one train variation per category; the later 3-var density gate did not promote ScienceWorld, but a future method with better validation or retrieval alignment could change that result.

Finally, we do not report provider dollar cost because it is unavailable in the final raw logs. Cost figures use average steps per task as a trajectory-cost proxy. We also lack a final retrieval-budget ablation, so this paper does not determine whether different CR/TR budgets would improve the mixed or negative cases.

nostic even after the denser 3-var KB gate.

6.5 Diagnostic Framework

Table 5 summarizes the explanatory framework suggested by the final audits. It should be read as a diagnostic hypothesis, not as a universal law.

The practical implication is conservative. Before deploying memory retrieval, practitioners should ask whether prior corrections are likely to transfer across tasks, whether the action/state representation preserves that transfer, and whether hard-negative controls separate relevant memory from irrelevant context. If those checks fail, intra-task self-correction, retrieval gating, or stronger validation may be preferable to simply adding more memories.

Table 5: Diagnostic framework for validated memory transfer in the final audited environments.

Property	ALFWorld	SQL	ScienceWorld
Task compositionality	High	Moderate	Moderate but diverse
Action-space regularity	High	High syntax, local grounding	Mixed procedural dynamics
Learning transfer radius	High	Narrow	Unreliable under current KB/retrieval
CR+TR vs. ReAct	+14.9pp	+0.5pp	−2.2pp
Hard-neg vs. ReAct	−12.7pp	−2.0pp	−1.1pp
Interpretation	Positive transfer	Narrow diagnostic	Diagnostic non-promotion

8 Conclusion

This work provides evidence that validated episodic memory is a conditional transfer mechanism, not an automatic improvement for LLM agents. In the final audited results, ALFWorld shows the clear positive case: CR+TR improves ReAct by +14.9pp and hard-negative retrieval substantially hurts. SQL shows only narrow transfer, with a +0.5pp CR+TR gain and negative CR/hard-negative results. ScienceWorld remains diagnostic negative evidence: neither the final 1-var KB nor the later 3-var density gate promotes CR+TR over ReAct and hard-negative controls.

We argue that task compositionality, action-space regularity, and learning transfer radius explain this pattern. These properties should be treated as diagnostic criteria rather than universal predictors. Future work should test additional models and domains, develop retrieval gates that estimate transferability before injecting memory, and improve validation signals so that denser memories become more useful rather than merely more numerous.

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Appendix

A Retrieval Algorithms

Algorithm 1 Context Retrieval from Knowledge Base

Require: Task description q ; knowledge base $\mathcal{K}$; neighbor count $k=5$; scaling factor $\alpha=1.5$; cap $p \in [5, 25]$; validation weights W

Ensure: Selected learnings $\mathcal{C}$

```

1:  $\mathbf{v}_q \leftarrow \text{EMBED}(q)$ 
2:  $\mathcal{T}_{\text{top}} \leftarrow \text{Top-}k \text{ tasks in } \mathcal{K} \text{ by } \text{sim}(\mathbf{v}_q, \mathbf{v}_t) \text{ \{filter exact matches\}}$ 
3:  $L_{\text{max}} \leftarrow \max_{t \in \mathcal{T}_{\text{top}}} \text{LEARNINGCOUNT}(t)$ 
4:  $N_{\text{pick}} \leftarrow \min(\lceil \alpha \cdot L_{\text{max}} \rceil, p)$ 
5:  $\mathcal{L}_{\text{pool}} \leftarrow \emptyset$ 
6: for each  $(t, s_t) \in \mathcal{T}_{\text{top}}$  do
7:   for each learning  $\ell \in \text{GETLEARNINGS}(t)$  do
8:      $\text{score}(\ell) \leftarrow s_t \times W[\ell.\text{valid\_level}]$ 
9:      $\mathcal{L}_{\text{pool}} \leftarrow \mathcal{L}_{\text{pool}} \cup \{(\ell, \text{score}(\ell))\}$ 
10:  end for
11: end for
12: Sort  $\mathcal{L}_{\text{pool}}$  by score descending; deduplicate by learning text
13:  $\mathcal{C} \leftarrow \text{first } N_{\text{pick}} \text{ elements of } \mathcal{L}_{\text{pool}}$ 
14: return  $\mathcal{C}$ 

```

B Knowledge Base Schema

Each entry in the knowledge base is a structured record containing the following fields:

- **Task metadata:** task description, object type, action verbs, goal phase (SEARCH, ACQUIRE, TRANSFORM, PLACE, or RECOVER).
- **Issue-learning pair:** `issue_text` (the specific mistake encountered) and `learning_text` (the successful correction).
- **Validation level:** a confidence label reflecting downstream outcome:

Algorithm 2 Hybrid Tool Retrieval

Require: Issue description I_q ; knowledge base $\mathcal{K}$; result count $k=3$; dense weight $\alpha=0.7$; sparse weight $\beta=0.3$; validation weights W

Ensure: Top- k issue-learning pairs

```

1:  $q' \leftarrow \text{CLEAN}(I_q) \text{ \{remove digits, normalize\}}$ 
2:  $\mathbf{v}_q \leftarrow \text{EMBED}(q')$ 
3:  $\mathbf{s}_{\text{cos}} \leftarrow \text{COSINESIM}(\mathbf{v}_q, \mathcal{K}.\text{issue\_embeddings})$ 
4:  $\mathbf{s}_{\text{bm25}} \leftarrow \text{BM25}(q', \mathcal{K}.\text{index})$ 
5:  $\mathcal{C} \leftarrow \text{Top}(\mathbf{s}_{\text{cos}}, 100) \cup \text{Top}(\mathbf{s}_{\text{bm25}}, 100) \text{ \{candidate pool\}}$ 
6: for each  $i \in \mathcal{C}$  do
7:    $c_i \leftarrow \text{clamp}(\mathbf{s}_{\text{cos}}[i], 0, 1)$ 
8:    $b_i \leftarrow \text{MINMAXNORM}(\mathbf{s}_{\text{bm25}}[i], \mathcal{C})$ 
9:    $S_{\text{hybrid}} \leftarrow \alpha \cdot c_i + \beta \cdot b_i$ 
10:   $\text{score}(i) \leftarrow S_{\text{hybrid}} \times W[\mathcal{K}[i].\text{valid\_level}]$ 
11: end for
12: Sort  $\mathcal{C}$  by score descending
13: return first  $k$  elements of  $\mathcal{C}$ 

```

- **VALID_NEXT_TRIAL:** the learning was applied and led to success in a subsequent trial (highest confidence). Weight $\omega_{\text{next}} = 1.0$.
- **VALID_SAME_TRIAL:** the learning is associated with success within the same trial. Weight $\omega_{\text{same}} = 0.9$.
- **CANDIDATE:** the learning was generated but the task was not completed (lowest confidence). Weight $\omega_{\text{cand}} = 0.6$.

- **Provenance:** references to the source task, trial number, and step range for both the issue and its resolution.

C Experimental Details

C.1 Dataset Statistics

ALFWorld. The full ALFWorld dataset (Shridhar et al., 2021) comprises 3,827 games split into a training set (3,553 games), a validation-seen set (140 games), and a validation-unseen set (134 games). The final evaluation uses the 134-task validation-unseen split. For knowledge base construction, we use a curated training subset with no validation overlap. The final audited KB contains 373 memories, including 101 VALID_NEXT_TRIAL entries.

SQL. The final SQL evaluation uses 200 held-out test tasks. The train-only sanitized KB covers

516 training tasks and contains 415 memories, including 92 VALID_NEXT_TRIAL entries. Protocol memories with dev/test leakage risk are quarantined before final evaluation.

ScienceWorld. The final ScienceWorld evaluation uses 30 held-out test categories from ScienceWorld (Wang et al., 2022). The final train-only KB covers all 30 categories with one train variation per category and contains 129 memories, including 34 VALID_NEXT_TRIAL entries. A separate Phase 4.1 diagnostic gate later tested a denser three-variation/category KB; it did not promote ScienceWorld to a headline positive result.

C.2 Framework Variants

We evaluate five inference-time framework variants in each final environment:

1. **ReAct**: the base agent with no memory (Yao et al., 2023), serving as the primary baseline.
2. **ReAct + CR**: base agent augmented with Context Retrieval only.
3. **ReAct + TR**: base agent augmented with Tool Retrieval only.
4. **ReAct + CR + TR**: base agent augmented with both retrieval mechanisms.
5. **ReAct + hard-neg CR + TR**: ablation control that retrieves maximally *irrelevant* entries via inverted retrieval (Section 3.3.4).

All final inference variants execute a single trial per held-out task. Reflexion is used to build the training KBs with up to 7 trials per training task, but it is not a headline final inference variant in Table 1.

C.3 Evaluation Metrics

Accuracy (success rate). The fraction of tasks completed successfully, computed as a binary outcome per task.

Average Steps per Task. The mean total thought-action-observation steps per held-out task. Because all final inference variants are single-trial, this is the trajectory-cost proxy used in Figure 5.

Gain from ReAct. The percentage-point difference between a variant’s accuracy and the ReAct baseline accuracy on the same split.

C.4 Hyperparameters

Table 6 lists all hyperparameters used across experiments.

Parameter	Value	Description
<i>Context Retrieval</i>		
p	[5, 25]	Tunable cap on retrieved learnings
α	1.5	Dynamic context sizing factor
k_{CR}	5	Neighbor tasks retrieved
<i>Tool Retrieval</i>		
k_{TR}	3	Results returned per query
Dense weight	0.7	Cosine similarity fusion weight
Sparse weight	0.3	BM25 fusion weight
Candidate pool	100+100	Top-100 from each strategy
<i>Validation Weights</i>		
ω_{next}	1.0	VALID_NEXT_TRIAL
ω_{same}	0.9	VALID_SAME_TRIAL
ω_{cand}	0.6	CANDIDATE
<i>Agent Configuration</i>		
Max steps/task	49	Episode length limit
Max valid actions	80	ScienceWorld action cap
Temperature	0.0	Deterministic decoding
Max output tokens	2,048	LLM generation limit
Max Reflexion trials	7	Retry budget

Table 6: Hyperparameters used across all experiments.

C.5 Implementation Details

All agent reasoning uses Gemini 2.5 Flash routed through the OpenRouter API. We use greedy decoding (temperature 0.0) with a maximum output length of 2,048 tokens. API calls employ exponential backoff (2 s–30 s) with up to 8 retry attempts.

Retrieval embeddings use gemini-embedding-001 via the Google GenAI SDK directly, with a batch size of 100 texts per API call. Embeddings are cached to disk alongside the knowledge base file; the cache includes a hash of the source file and the embedding model name for automatic invalidation on changes.

Each final held-out task trial is limited to 49 thought-action-observation steps across all agent variants. ScienceWorld additionally uses max_valid_actions=80. Reflexion training trajectories allow up to 7 trials per task for KB construction.

D Diagnostic Gates Not Promoted to Headline Results

ScienceWorld density gate. The Phase 4.1 gate rebuilt the ScienceWorld KB with three train variations per category and reran all 30 held-out test categories across seeds 0, 1, and 2. The gate completed, but did not promote ScienceWorld: CR+TR underperformed same-run ReAct on mean success and mean reward, and hard-negative CR+TR matched or beat CR+TR on success in every seed. This supports the paper’s conservative interpretation that density alone is insufficient.

TextWorld pilot. The Phase 4.2 TextWorld pilot was pre-registered as an exploratory environment-selection gate, not as a final paper result. It used 12 generated held-out ‘.ulx’ games. ReAct solved 10/12, CR+TR solved 11/12, and hard-negative CR+TR solved 10/12. The gate failed because the ReAct baseline was near ceiling and CR+TR did not beat ReAct or hard-negative retrieval by the pre-registered two-task margin. We therefore do not include TextWorld as a headline result.